

Stochastic Programming.
Problems session 5

19. A company wants to contract workers of two profiles to start a project. The company needs 10 workers: x_1 and x_2 of profiles 1 and 2, respectively. To simplify, we will assume that x_1 and x_2 are continuous variables. The costs for each worker is 5 for profile 1 and 6 for profile 2, in the first stage.

The number of needed workers of each profile is ξ_1 and ξ_2 , and we know that $\xi_1 + \xi_2 = 10$. Changing workers from profile 1 to 2, once started the project, has a cost of 4 for each worker; and changing workers from profile 2 to 1 has a cost of 2 for each worker. Variable ξ_1 is continuous uniform in interval $[4, 8]$. We formulate the following problem:

$$\begin{array}{ll} \min & 5x_1 + 6x_2 + E_{\xi_1}[Q(x_1, x_2, \xi_1)] \\ \text{s.to} & x_1 + x_2 = 10 \\ & x_1, x_2 \geq 0 \end{array} \quad \text{where} \quad Q(x_1, x_2, \xi_1) = \begin{cases} 2(\xi_1 - x_1) & \text{if } x_1 \leq \xi_1 \\ 4(x_1 - \xi_1) & \text{if } x_1 \geq \xi_1. \end{cases}$$

- (a) Compute the solution of the above problem. Hint: Take the equality constraint of the first stage to remove x_2 from the formulation.
- (b) Compute EVPI and VSS.
20. To overcome the economic crisis a country decides to establish direct subsidies (giving nonrepayable money) to encourage consumption. The total budget for the subsidies is x , between 3 and 6 million €. It is estimated that each € in direct grants will mean ξ € to the country economy, where ξ is assumed continuous uniform between 1 and 2.

If this policy ξx does not provide at least 6 million € of benefit to the country economy, we will assume that the country incurs in a large and fixed cost of f millions €. Otherwise, if this policy provides at least 6 million, there isn't any cost of second stage.

The country formulates the following stochastic problem:

$$\begin{array}{ll} \min & x + E_{\xi}[Q(x, \xi)] \\ \text{s.to} & 3 \leq x \leq 6 \end{array} \quad \text{where} \quad Q(x, \xi) = \begin{cases} f & \text{if } x \leq \xi x < 6 \\ 0 & \text{if } \xi x \geq 6 \end{cases}$$

- (a) Compute the optimal number x of million € as a function of f .
- (b) Compute EVPI, for $f = 6$.
21. The company "ApplePie" wants to launch a new computer gadget, and it must decide the selling price (x €/unit). The selling price will be between minimum and maximum prices l and u . Demand ξ follows a continuous uniform distribution between m/x and $3m/x$, m being a parameter of the problem. Clearly, demand depends on the selling price, and if in the future demand is below t , t being a parameter between m/x and $3m/x$, then the company will have a risk rx^2 (where r is a risk factor). The company formulates the following stochastic problem:

$$\begin{array}{ll} \min & -\beta x + E_{\xi}[Q(x, \xi)] \\ \text{s.to} & l \leq x \leq u \end{array} \quad \text{where} \quad Q(x, \xi) = \begin{cases} rx^2 & \text{if } \xi < t \\ 0 & \text{if } \xi \geq t, \end{cases}$$

where β is related to the selling benefits.

- (a) Compute the optimal selling price x . Take: $l = 5, u = 20, \beta = 50, r = 10, t = \frac{3m}{2x}$.
- (b) Compute VSS.

22. John Thedoor, president of a prestigious football team from Barcelona wants to change some of the players. He considers to buy up to n new players, between 2 and 5. The cost of every new player is $c \text{ €}$. Every new player provides an ability to win the cup next season. The total ability that will be needed to win the cup is a random variable ξ that follows a continuous uniform distribution in $[6, 10]$. The club formulates the next stochastic problem to decide how many players buy up:

$$\begin{array}{ll} \min & cx + E_{\xi}[Q(x, \xi)] \\ \text{s.to} & 2 \leq x \leq 5 \end{array} \quad \text{where} \quad Q(x, \xi) = \begin{array}{ll} \min & y \\ \text{s.to} & 2x + y \geq \xi \\ & y \geq 0 \end{array}$$

- (a) Compute the number of players to buy up, in terms of c . Consider that all variables are continuous.
- (b) Compute the feasible sets K_1 and K_2 .
- (c) Compute EVPI, assuming that $c = 2$.